Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide K_4 Fe (CN)_6. After filtering, the precipitate is dried and weighed. The reactions which occur are ZnS + 2HCl \rightarrow ZnCl_2 + H_2 S 2ZnCl_2 + K_4 Fe (CN)_6 \rightarrow Zn_2 Fe (CN)_6 + 4 KCl If a 2 gram sample of rock yields 0.969 gram of Zn_2Fe(CN)_6, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight Zn_2 Fe (CN)_6 = 342.6.

Answer Choices:

A. 10 % Zn
B. 22 % Zn
C. 20 % Zn
D. 19 % Zn
E. 15 % Zn
F. 18 % Zn
G. 16 % Zn
H. 14 % Zn
I. 12 % Zn
J. 25 % Zn

Reasoning + Answer:

</tool_call> reasoning process here

The problem involves determining the percentage of zinc (Zn) in a rock sample based on the mass of a precipitate formed in a chemical analysis.

Given:

- Mass of rock sample = 2 grams
- Mass of precipitate (ZnâĤĤFe(CN)âĤĨ) obtained = 0.969 grams
- Molar mass of ZnâĤĤFe(CN)âĤĨ = 342.6 g/mol
- Molar mass of Zn = 65.4 g/mol âĨĴ so 2 atoms of Zn = 2 ÃĹ 65.4 = 130.8 g/mol

Step 1: Find the mass of Zn in the precipitate.

From the molar mass of ZnâĤĤFe(CN)âĤĨ, the mass of Zn in one mole is 130.8 g.

So, fraction of Zn in ZnâĤĤFe(CN)âĤĨ = (130.8 / 342.6)

Mass of Zn in 0.969 g of ZnâĤĤFe(CN)âĤĨ =

0.969 ÃĹ (130.8 / 342.6)

Calculate that:

130.8 / 342.6 âīĪ 0.3816

0.969 ÃĹ 0.3816 âīĪ 0.3700 g of Zn

Step 2: This mass of Zn came from the 2 g sample, so the percentage of Zn in the rock is:

(0.3700 / 2) ÃĹ 100 = 18.5% âīĪ 18%

So the closest answer is 18%.

<answer>F</answer>